



MANIPAL SCHOOL OF INFORMATION SCIENCES

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(A constituent unit of MAHE, Manipal)

Master of Engineering – M.E in Computer Science and Engineering

Course Plan

Course Name	:	Real Time Operating Systems Lab
Course Code	:	ESE 5154
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Mr. Shreyas N
Name of the Program Coordinator	:	Dr. Sathyendranath Malli

Date:	01.08.2026
Signature of Program Coordinator	
with Date	

Date:	30.07.2026
Signature of Course Coordinator	
with Date	



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

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Table of Contents	
1 Course Plan	5
1.1 Primary Information	5
1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping	5
1.3 Assessment Plan	6
1.4 Lesson Plan	7
1.5 References	9
1.6 Other Resources (Online, Text, Multimedia, etc.)	9
1.7 Course Timetable	10
1.8 Assessment Plan	11
1.9 Assessment Details	13
1.10 Course Articulation Matrix	13

Program Education Objectives (PEOs)



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering – M.E in Computer Science and Engineering, programme are as follows

PEO No.	Education Objective
PEO 1	Enable to draw upon fundamental and advanced knowledge to apply analytical and computational approaches to solve technological problems in computer science.
PEO 2	Introduce state of art technologies in computer science and inculcate ethical practices to make industry-ready professionals.
PEO 3	Promote scientific and societal advancement through research and entrepreneurship.

Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering – M.E in Computer Science and Engineering, graduates will be able to:



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PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop software based on theoretical principles and practical knowledge.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analysing and decision-making.

1. Course Plan

1.1 Primary Information

Course Name	:	Real Time Operating Systems Lab [ESE 5154]
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L-T-P-C	:	0-0-3-1
Contact Hours	:	36 Hours
Pre-requisite	:	C Programming
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No. of Contact Hours	Program Outcomes (POs)	BL
CO1	Implement C programs to efficiently solve problems using multiple processes.	15	P03	3
CO2	Execute C programs for process scheduling and deadlock avoidance algorithms.	6	P04	3
CO3	Design multithreaded programs to solve problems considering process scheduling, deadlocks, or process synchronization on a given scenario.	15	P05	6

1.3 Assessment Plan

Components	Lab test	Flexible assessments	End semester exam (ESE) / Makeup exam
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Duration	90 minutes	2 weeks	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Apply	Apply, create	Apply, create
Pattern	Answer all the questions	Implement process, system calls to solve the given problems	Answer all the questions
Schedule	As per academic calendar.	Assignment submission: sept30th, october 15th	As per academic calendar.
Topics covered	process, system calls, scheduling algorithms	process, system calls, scheduling algorithms, FREERTOS	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.

1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---



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L1	Conduct experiment to display the process states of various active process.	C01
L2	Conduct experiment to display the process states of various active process.	C01
L3	Conduct experiment to display the process states of various active process.	C01
L4	Conduct experiments by using POSIX system calls.	C01
L5	Conduct experiments by using POSIX system calls.	C01
L6	Conduct experiment on process scheduling algorithms.	C02
L7	Conduct experiment to demonstrate process deadlocks.	C02
L8	Conduct experiment to demonstrate process synchronization.	C03
L9	Conduct experiment to demonstrate process synchronization.	C03
L10	Conduct Experiment on multithread programming.	C03
L11	Conduct Experiment on multithread programming.	C03
L12	Conduct experiment to demonstrate the working of real-time operating systems using FREERTOS.	C03
L13	MSC	C01, C03



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1.5 References



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1. "Operating System principles", Seventh Edition, Abraham Silberschatz, Peter Galvvin, Grag Gagne. John Wiley Publications.
2. "Real – Time Systems and Programming Languages", Allan Burns, Andy Wellings.
3. "Operating Stems Concepts and Design", Milan Milenkovic.
4. "Design of Unix Operating System", Maurice Bach (IPC).
5. "The C Programming Language", Kerningham & Ritchie.

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.

1.7 Course Timetable



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Semester - I						Lab:	Embedded Systems Lab		
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday									
Tuesday									
Wednesday									
Thursday									
Friday									
Saturday									

RTOS Lab

1.8 Assessment Plan

COs	Marks & Weightage
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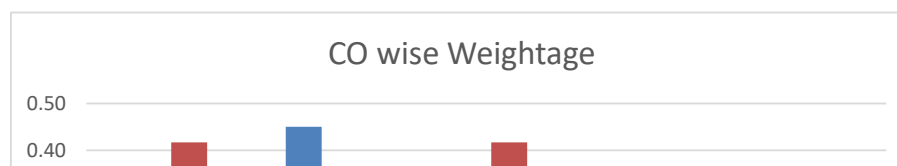


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CO No.	CO Name	Lab test marks (Max. 30)	Assignment Marks (Max. 20)	ESE Marks (Max. 50)	Actual CO wise Weightage	Expected CO wise Weightage
CO1	Implement C programs to efficiently solve problems using multiple processes.	15	5	10	0.30	0.42
CO2	Execute C programs for process scheduling and deadlock avoidance algorithms.		10	20	0.45	0.17
CO3	Design multithreaded programs to solve problems considering process scheduling, deadlocks, or process synchronization on a given scenario.	15	5	20	0.25	0.42
	Marks (weightage)	0.3	0.2	0.5	1.00	1.00

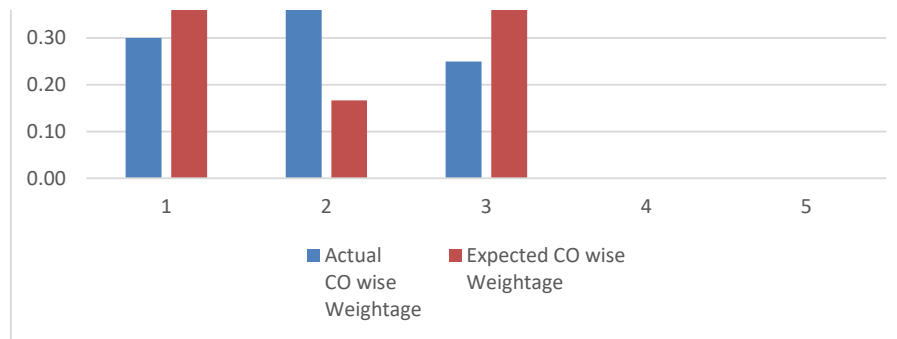




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1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

SI. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	Lab test	0.3	1	- Performance is measured using lab test attainment level. - Reference: question paper and answer scheme. - The lab test is assessed for a maximum of 30 marks.



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2	Assignments	0.2	1	• Performance is measured using assignments attainment level. • Assignment is evaluated for a maximum of 20 marks.
3	ESE	0.5	1	• Performance is measured using ESE attainment level. • Reference: question paper and answer scheme. • ESE is assessed for a maximum of 50 marks.

1.10 Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5
CO1			Y		
CO2				Y	
CO3					Y
Average Articulation Level			*	*	*